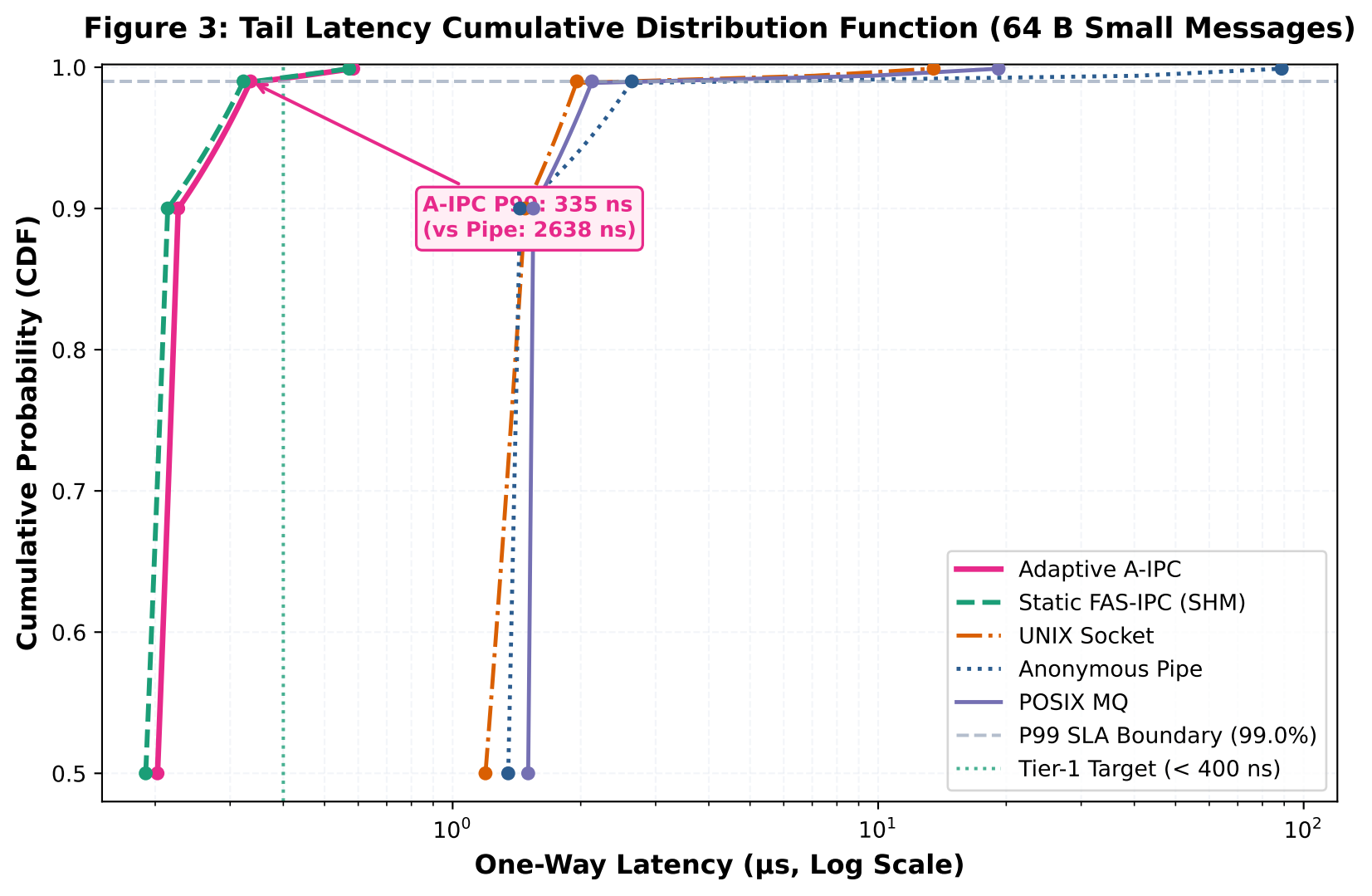
Figure 3: Tail Latency Cumulative Distribution Function (64 B Small Messages)
Cumulative Probability (CDF)
One-Way Latency (µs, Log Scale)
1.0
0.9
0.8
0.7
0.6
0.5
10^0
10^1
10^2
A-IPC P99: 335 ns
(vs Pipe: 2638 ns)
Adaptive A-IPC
Static FAS-IPC (SHM)
UNIX Socket
Anonymous Pipe
POSIX MQ
P99 SLA Boundary (99.0%)
Tier-1 Target (< 400 ns)